The relationship between steroid treatment and mortality in patients with COVID-19 followed up in an intensive care unit

Huseyin Ali Ozturk and Fatih Necip Arici

Department of Internal Medicine, University of Health Sciences—Adana Health Practice and Research Center, Adana, Turkey

ABSTRACT

Aim. Optimal treatment of the coronavirus disease (COVID-19) is still unclear. It has been reported that the use of different doses of corticosteroid treatments may reduce mortality. In our study, we aimed to find the effect of corticosteroid treatment dose on mortality of patients followed up in intensive care due to COVID-19.

Methods. Our retrospective, descriptive and single-centre study included 102 patients diagnosed with COVID-19 who were followed up in intensive care unit, 28 of whom received pulse steroids and 74 of whom received high dose steroids. Laboratory values, duration of intensive care unit and mortality rates of the patients were evaluated.

Results. Mortality was found to be statistically significantly lower in the group receiving pulse steroid compared to the group receiving high dose steroid. In multivariate logistic regression analysis, age and pulse steroid were found to be independent predictors of mortality. According to this analysis, each 10-year increase in age increased mortality by 4.8%, whereas pulse steroid decreased mortality by 79.4%.

Conclusion. In our study, we found that mortality was statistically significantly lower in the group of patients receiving pulse steroids than in the group receiving high dose steroids. We found that the number of patients using pulse steroids was statistically significantly lower in the group with mortality. We found that age and pulse steroid independently determined the patients with mortality.

Subjects Emergency and Critical Care, Internal Medicine, Public Health, COVID-19

Keywords COVID-19, Mortality, Pulse methylprednisolone, Intensive care

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Corresponding author
Huseyin Ali Ozturk,
drozturkhuseyinali@gmail.com

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INTRODUCTION

Coronavirus disease (COVID-19) is an infectious disease caused by a newly discovered coronavirus (SARS-CoV-2). The average incubation period is about 4 days and about 98% of those affected show symptoms within 11.5 days. Clinical manifestations may range from asymptomatic to pneumonia or acute respiratory distress syndrome (ARDS). This disease, which affects millions of people worldwide, is known to cause higher complications, mortality and morbidity, especially in older people and people with known hypertension, diabetes and obesity (*Guan et al., in press*; *Sumbul et al., 2021*; *Koc et al., 2020*). Disease severity can be clinically classified as mild, moderate, severe and critical (*Wu & McGoogan, 2020*). The treatment strategy of COVID-19 treatment with optimal efficacy and safety

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is still unclear. There are reports in the literature that a clinical picture associated with cytokine release syndrome in the course of COVID-19 disease can have serious and fatal consequences. Many authors have emphasised the importance of early recognition of the hyperinflammatory state resulting from a much more exaggerated and uncontrolled inflammatory response to SARS CoV-2. Due to the global spread of the pandemic, it has been reported that the use of different doses of corticosteroid treatments may reduce ARDS severity and mortality (*Chaudhuri et al., 2021*; *Angus et al., 2020*). Therefore, accurate and timely identification of the subgroup in which the type of corticosteroid treatment may reduce ARDS and mortality is important for planning an effective and safe treatment. In our study, we aimed to find the effect of corticosteroid treatment dose on mortality of patients followed up in intensive care due to COVID-19.

MATERIALS AND METHOD

Study population

Our study was designed as a retrospective, descriptive and single-center study. Between 01.05.2020–01.05.2022, patients who were diagnosed with COVID-19 at Adana City Training and Research Hospital, who received high-dose steroids or pulse steroids and had a history of adult intensive care unit hospitalisation were included. Polymerase chain reaction (PCR) positivity for SARS-CoV-2 and typical radiological findings (bilateral ground-glass areas) were the criteria for the diagnosis of COVID-19. All patients included in the study were moderately-serious patients who were hospitalised in intensive care units and needed oxygen support. Patients with similar age, gender and body mass index (BMI) were selected for the study groups. The study data were obtained from the data processing records of Adana City Training and Research Hospital and daily visit files of the patients. Detailed anamnesis and physical examination were performed in all patients; age, gender, medical history, physical examination findings and laboratory measurements were recorded in their files performed. Adana City Training and Research Hospital Ethics Committee approved the study with decision number 2784 dated 17.08.2023. Smoking history, whether the patients were receiving any oral or inhaled corticosteroid treatment were recorded. Lung tomographies of the patients were evaluated. Duration of intensive care unit stay, APACHE 2 scores, duration of mechanical ventilator stay, peep/fio2 (positive end-expiratory pressure/fraction of inspired oxygen) ratios and mortality rates were evaluated.

Corticosteroid treatment was categorised as pulse corticosteroid and high dose corticosteroid treatment. High dose corticosteroid treatment was defined as 1 mg/kg methylprednisolone or equivalent dose of dexamethasone daily. Pulse corticosteroid was defined as 500 mg methylprednisolone. The duration of pulse steroid treatment was 3 days. The study was conducted in accordance with the Declaration of Helsinki.

Laboratory measurements

Laboratory procedures of the study were performed in the Biochemistry Laboratory of Health Sciences University Adana City Training and Research Hospital. Laboratory

parameters of the patients were measured with an automated chemistry analyser (Abbott Aeroset) using appropriate commercial kits (Abbott, Abbott Park, IL, USA).

Statistical analysis

All analyses were performed using the statistical software package SPSS 24.0 (Chicago, IL, USA). Student t-test or Mann–Whitney U was used for the comparison of continuous variables between groups. The chi-square ($\chi 2$) test was used to compare categorical variables. In the univariate analysis for the independent determination of mortality, multivariate logistic regression analysis was performed with statistically significant parameters with p value < 0.05. The sensitivity and specificity of the mortality predictive values were performed by ROC curve analysis. Statistical significance was accepted as $p < 0.05$ for all comparisons.

RESULTS

Patients with similar age and comorbidities were included in our study. Patients were divided into two groups as pulse steroid recipients (group 1) and high dose steroid recipients (group 2). Group 1 consisted of 28 patients, 20 males and eight females. In group 2, there were 74 patients, 45 males and 29 females. No statistically significant difference was found in terms of chronic obstructive pulmonary disease, diabetes, hypertension, heart failure, chronic kidney disease, pulse rate, systolic blood pressure, diastolic blood pressure, fever and body mass index. When the patient groups were evaluated in terms of biochemical parameters, urea level was found to be statistically significantly higher in group 1 than in group 2. No significant difference was observed when the groups were compared in terms of days of intensive care unit hospitalisation, APACHE-2 score and days on mechanical ventilator. When the groups were compared in terms of mortality, mortality rate was significantly lower in group 1 (Table 1).

The patients included in the study were divided into two groups according to mortality. Age was found to be statistically significantly higher and the number of patients using pulse steroid was found to be significantly lower in the group with mortality than in the group without mortality. When the groups were compared in terms of laboratory values, no significant difference was observed between both groups. When the groups were compared in terms of intensive care follow-up, APACHE-2 score was found to be statistically significantly higher in the group with mortality (Table 2).

In multivariate logistic regression analysis, age and pulse steroid use were found to be independent predictors of mortality. According to this analysis, each 10-year increase in age increased mortality by 4.8%, whereas pulse steroid use decreased mortality by 79.4% (Table 3).

ROC analysis was performed with the parameters found to be significant in multivariate logistic regression analysis. When the cut-off value of age was taken as 60.5, it was found to be 67.6% sensitive and 64.6% specific in determining mortality (Table 4, Fig. 1).

Table 1 **Evaluation of the study groups in terms of demographic, clinical characteristics, laboratory values and intensive care follow-up.** Statistically significant results are in bold.

Variables	Group 1 *n* = 28	Group 2 *n* = 74	*p*
Age (years)	61.3 ± 14.6	60.1 ± 14.8	0.711
Gender, (M/F), *n*	20/8	45/29	0.312
COPD, *n*	3 (10.7%)	5 (6.8%)	0.626
CKD, *n*	6 (21.4%)	6 (8.1%)	0.168
DM, *n*	12 (42.9%)	19 (25.7%)	0.185
HT, *n*	15 (53.6%)	24 (32.4%)	0.133
HF, *n*	4 (14.3%)	5 (6.8%)	0.332
Pulse (pulse/minute)	86.6 ± 21.7	90.5 ± 17.8	0.367
SBP (mmHg)	130.5 ± 23.4	127.4 ± 22.5	0.560
DBP (mmHg)	75.5 ± 14.3	73.9 ± 12.6	0.595
Fever (°C)	36.3 ± 0.41	36.3 ± 0.29	0.930
BMI (kg/m2)	28.23 ± 3.50	28.1 ± 4.94	0.929
Troponin (ng/l)	242.2 ± 792.2	60.7 ± 201.1	0.278
Glucose (mg/dl)	182.7 ± 84.0	191.5 ± 93.9	0.664
Urea (mg/dl)	78.5 ± 53.6	54.3 ± 38.0	**0.035**
Creatinine (mg/dl)	1.96 ± 2.4	1.00 ± 0.75	0.052
Sodium (mmol/l)	135.8 ± 4.94	137.2 ± 4.41	0.174
Potassium (mmol/l)	4.27 ± 0.43	4.16 ± 0.62	0.424
AST (U/L)	45.5 ± 24.0	44.0 ± 32.8	0.833
ALT (U/L)	47.1 ± 27.4	43.4 ± 34.2	0.576
LDH (U/L)	540.9 ± 190.9	491.8 ± 210.6	0.284
Calcium (mg/dl)	8.32 ± 0.60	8.24 ± 0.60	0.537
Ferritin (pg/ml)	854.9 ± 882.5	565.9 ± 760.1	0.105
CRP (mg/L)	126.0 ± 88.7	130.5 ± 96.8	0.830
WBC (10 3/pl)	12.7 ± 5.72	10.3 ± 6.49	0.094
RBC (10 6/pl)	4.50 ± 0.69	4.28 ± 0.80	0.206
Hemoglobin (g/dl)	13.0 ± 2.25	12.2 ± 2.07	0.109
Neutrophil (10 3/pl)	11.18 ± 5.02	9.12 ± 6.01	0.110
Lymphocyte (10 3/pl)	0.69 ± 0.54	0.64 ± 0.50	0.674
INR, *n*	1.32 ± 1.76	1.02 ± 0.33	0.371
D-dimer (pg/ml)	2449.2 ± 3087.7	1929.2 ± 2941.7	0.434
Fibrinogen (mg/dl)	622.2 ± 138.7	604.6 ± 162.8	0.613
Pro-BNP (pg/ml)	6023.4 ± 9905.7	2048.8 ± 3588.1	0.102
Triglyceride (mg/dl)	201.7 ± 104.9	167.1 ± 78.0	0.246
LDL (mg/dl)	100.2 ± 34.8	118.8 ± 35.3	0.199
Days of intensive care hospitalisation, n	14.6 ± 10.1	15.8 ± 15.6	0.712
PF Rate	159.0 ± 37.7	178.3 ± 46.2	0.052
APACHE-2 score	21.1 ± 12.4	18.36 ± 10.31	0.269

(*continued on next page*)

Table 1 (*continued*)

Variables	Group 1 $n = 28$	Group 2 $n = 74$	p
MV Days, n	4.92 ± 8.40	5.05 ± 12.0	0.959
Mortality, n	8 (21.4%)	31 (41.9%)	**0.041**

Notes.

COPD, Chronic obstructive pulmonary disease; CKD, Chronic kidney disease; DM, Diabetes mellitus; HT, Hypertension; HF, Heart failure; SDB, Systolic blood pressure; DBP, Diastolic blood pressure; BMI, Body mass index; AST, Aspartate aminotransferase; ALT, Alanine aminotransferase; LDH, Lactate dehydrogenase; CRP, C-reactive protein; WBC, White blood cell; RBC, Red blood cell; INR, International Normalised Ratio; Pro-BNP, Pro natriuretic peptide; LDL, Low density lipoprotein; PF Rate, Peep/Fio^2 rate; APACHE-2, Acute Physiology and Chronic Health Evaluation; MV, Mechanical ventilator.

DISCUSSION

Acute Respiratory Distress Syndrome (ARDS) related to COVID-19 presents multifaceted challenges, stemming from hyper-inflammatory responses and underlying pathophysiological conditions. Our study aimed to investigate the efficacy of pulse *versus* high-dose corticosteroid treatments in influencing mortality rates among critically ill COVID-19 patients. The findings indicate that pulse steroid therapy is associated with a significantly lower mortality rate compared to high-dose steroid therapy, suggesting that treatment strategies tailored to mitigate inflammation may prove beneficial in this patient population. Previous research has established a robust link between systemic inflammation and ARDS progression, emphasizing corticosteroids' role in attenuating this inflammatory response and improving outcomes (*Meyer, Gattinoni & Calfee, 2021*; *Chang et al., 2022*). The use of dexamethasone or other corticosteroids at equivalent doses is recommended for severe patients with COVID-19 receiving oxygen or respiratory support (*RECOVERY Collaborative Group et al., 2021*). The recommendation for corticosteroid use in severe COVID-19 cases aligns with our findings, reinforcing the theory that optimized steroid regimens can lead to improved survival rates (*RECOVERY Collaborative Group et al., 2021*; *Chaudhuri et al., 2021*). Specifically, the notable reduction in mortality associated with pulse steroid administration highlights the potential for lower-dose, intermittent steroid treatment to provide effective immunomodulation without the adverse effects often linked to continuous high-dose therapy. In our study, we found that mortality was statistically significantly lower in the group of patients receiving pulse steroids than in the group receiving high dose steroids. We found that age and pulse steroid use independently determined the patients with mortality.

In the study by *Choi et al. (2023)* patient groups with COVID-19 receiving pulse steroid and high dose steroid were compared. Mortality was found to be lower in the pulse steroid group. In addition, the duration of hospitalisation was shorter in the pulse steroid group. In the study by *Okano et al. (2022)* pulse steroid and high dose steroid groups were compared. In-hospital mortality and 28-day mortality were found to be lower in the group receiving pulse steroid. In another study by *Okano, Sakurai & Yamazaki (2023)*, pulse steroid treatment was found to reduce 28-day mortality in intensive care unit patients compared to high-dose steroid therapy. In the study by *Pinzón et al. (2021)*, it was found that recovery time and mortality were lower in the group receiving pulse steroid in intensive care unit compared to the group receiving high dose steroid. In our study, we found that

Table 2 **Evaluation of the groups in terms of demographic, clinical characteristics, laboratory values and intensive care follow-up according to mortality.** Statistically significant results are in bold.

Variables	With mortality $n = 37$	Without mortality $n = 65$	p
Age (years)	67.1 ± 13.8	56.7 ± 13.9	**<0.01**
Gender, (M/F), *n*	18/19	19/46	0.058
COPD, *n*	5 (13.5%)	3 (4.6%)	0.166
CKD, *n*	4 (10.8%)	8 (12.3%)	0.768
DM, *n*	13 (35.1%)	18 (27.7%)	0.464
HT, *n*	18 (48.6%)	21 (32.3%)	0.121
HF, *n*	5 (13.5%)	4 (6.2%)	0.275
Pulse steroid use, *n*	6 (16.2%)	22 (33.8%)	**0.042**
Pulse (pulse/minute)	91.1 ± 16.6	88.5 ± 20.2	0.509
SBP (mmHg)	130.3 ± 24.0	127.0 ± 22.0	0.488
DBP (mmHg)	71.1 ± 14.4	76.2 ± 14.06	0.53
Fever (°C)	36.3 ± 0.42	36.3 ± 0.26	0.764
BMI (kg/m^2)	27.7 ± 5.34	28.4 ± 4.07	0.447
Troponin (ng/l)	85.0 ± 268.4	118.3 ± 509.6	0.715
Glucose (mg/dl)	188.9 ± 95.7	189.2 ± 88.9	0.988
Urea (mg/dl)	62.5 ± 44.7	59.9 ± 43.8	0.761
Creatinine (mg/dl)	1.17 ± 0.87	1.31 ± 1.74	0.638
Sodium (mmol/l)	136.5 ± 5.38	137.0 ± 4.09	0.626
Potassium (mmol/l)	4.22 ± 0.60	4.18 ± 0.56	0.712
AST (U/L)	39.8 ± 27.0	47.0 ± 32.3	0.255
ALT (U/L)	36.3 ± 21.8	49.0 ± 36.4	0.58
LDH (U/L)	481.6 ± 184.1	518.8 ± 217.1	0.382
Calcium (mg/dl)	8.3 ± 0.62	8.25 ± 0.58	0.692
Ferritin (pg/ml)	701.7 ± 904.7	613.1 ± 742.1	0.594
CRP (mg/L)	147.9 ± 84.8	118.7 ± 98.3	0.134
WBC (10^3/pl)	11.2 ± 7.94	10.9 ± 5.3	0.809
RBC (10^6/pl)	4.21 ± 0.63	4.42 ± 0.84	0.204
Haemoglobin (g/dl)	12.1 ± 1.89	12.6 ± 2.26	0.236
Neutrophils (10^3/pl)	9.9 ± 7.4	9.5 ± 4.7	0.811
Lymphocytes (10^3/pl)	0.76 ± 0.60	0.59 ± 0.44	0.122
INR, *n*	1.34 ± 0.97	0.97 ± 0.11	0.158
D-dimer (pg/ml)	2139 ± 3514.0	2033.8 ± 2651.2	0.865
Fibrinogen (mg/dl)	628.6 ± 140.8	598.4 ± 164.3	0.351
Pro-BNP (pg/ml)	3027.2 ± 5650.5	2938.3 ± 6488.6	0.948
Triglycerides (mg/dl)	197.4 ± 117.7	176 ± 77.5	0.499
LDL (mg/dl)	119.7 ± 44.1	105.0 ± 32.2	0.367
Days of intensive care hospitalisation, *n*	15.3 ± 7.43	15.5 ± 17.1	0.949
PF Rate, *n*	165.6 ± 49.3	177.0 ± 41.7	0.221

(*continued on next page*)

Table 2 (*continued*)

Variables	With mortality $n = 37$	Without mortality $n = 65$	p
APACHE-2 score	23.3 ± 10.5	16.6 ± 10.5	**0.004**
MV Days, *n*	5.91 ± 5.38	4.48 ± 13.4	0.537

Notes.
COPD, Chronic obstructive pulmonary disease; CKD, Chronic kidney disease; DM, Diabetes mellitus; HT, Hypertension; HF, Heart failure; SDB, Systolic blood pressure; DBP, Diastolic blood pressure; BMI, Body mass index; AST, Aspartate aminotransferase; ALT, Alanine aminotransferase; LDH, Lactate dehydrogenase; CRP, C-reactive protein; WBC, White blood cell; RBC, Red blood cell; INR, International Normalised Ratio; Pro-BNP, Pro natriuretic peptide; LDL, Low density lipoprotein; PF Rate, Peep/Fio^2 rate; APACHE-2, Acute Physiology and Chronic Health Evaluation; MV, Mechanical ventilator.

Table 3 Multivariate logistic regression analysis for the detection of mortality. Statistically significant results are in bold.

Variables	Odd ratio	95% confidence interval	p
Age (years)	1.048	1.008–1.089	**0.019**
Pulse steroid use	0.206	0.062–0.678	**0.009**
APACHE-2 Score	1.049	1.000–1.102	0.052

Notes.
APACHE-2, Acute Physiology and Chronic Health Evaluation.

Table 4 Age-standardised roc analysis for the detection of mortality.

Variable	Area under the ROC curve	p	Cut-off	Sensitivity	Specificity
Age	0.708 (0.603–0.813)	0.001	60.5	67.6%	64.6%

mortality was lower in the group receiving pulse steroid. The study groups were similar in terms of age, comorbidities and laboratory values. The peep/fio2 ratio and APACHE-2 scores, which are parameters related to oxygen demand, were similar between both groups. For these reasons, it is important that we found lower mortality in the group receiving pulse steroid. In addition, while the group with high mortality and the group with low mortality were similar in terms of comorbidities and laboratory values, the low number of pulse steroid users in the group with high mortality is important. In multivariate logistic regression analysis, we found that pulse steroid use reduced mortality by 79.4%. Considering these, pulse steroid treatment may be preferred in appropriate patient groups.

It is essential to acknowledge certain limitations in our study, including the small sample size compared to larger published trials, which may influence the generalizability of our findings. Additionally, the predominant use of methylprednisolone in our cohort, as opposed to dexamethasone employed in other studies, could contribute to variances observed in mortality rates and other outcomes. In the study of *Batirel et al. (2021)* patients were divided into 3 groups as standard care treatment, high dose steroid treatment and pulse-steroid treatment. In the results of the study, shorter intensive care unit stay was observed in the pulse steroid group. Mortality was similar between the two groups receiving steroid treatment (*Batirel et al., 2021*). In the study by *Gundogdu et al. (2021)* patients were divided into three groups as control group, high dose and pulse steroid treatment group. As

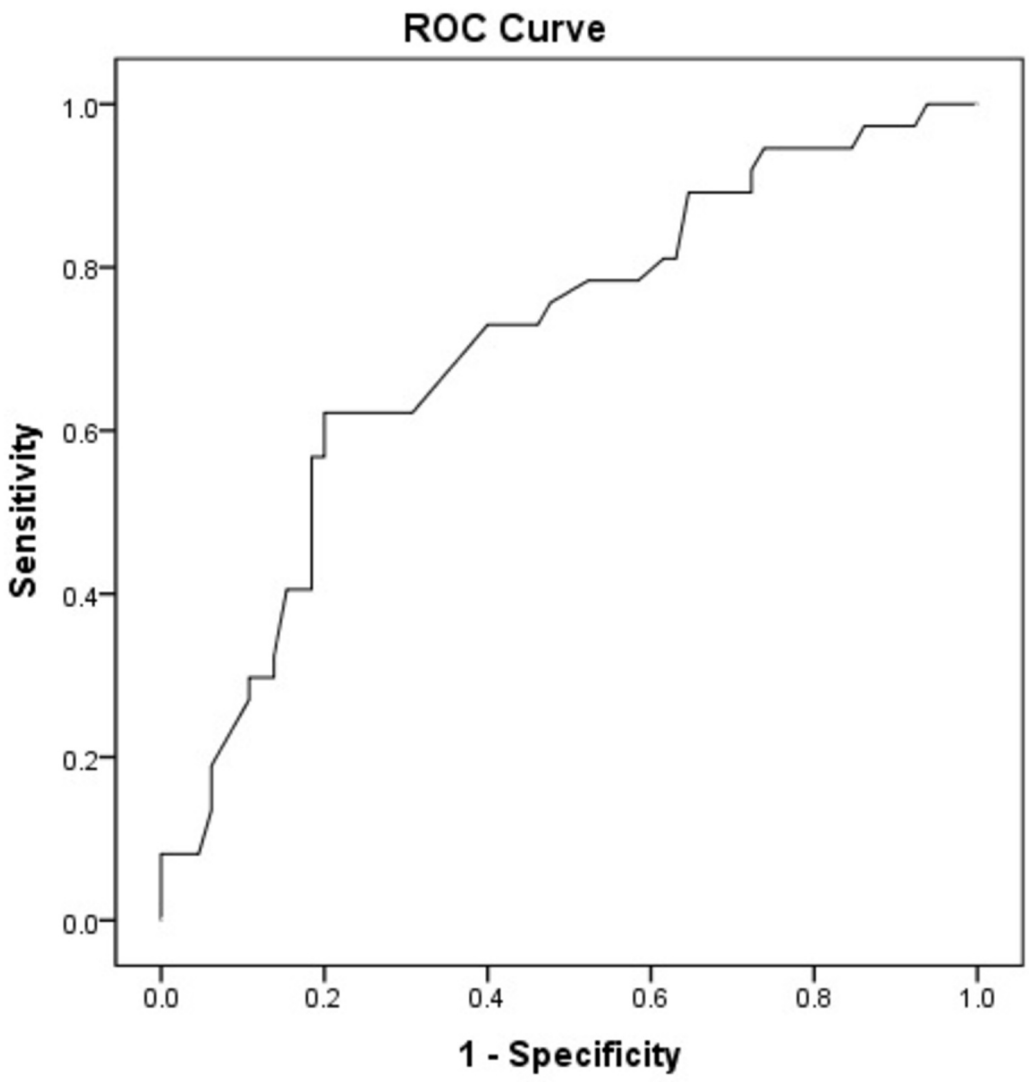

Figure 1 **ROC curve of age identifying patients with mortality in patients with Covid-19.**
Full-size DOI: 10.7717/peerj.18606/fig-1

a result of the study, it was found that pulse steroid treatment did not shorten the duration of hospital stay and did not reduce the need for intubation. In addition, similar mortality rates were found in both groups receiving steroid treatment (*Gundogdu et al., 2021*). In the study by *Corral-Gudino et al. (2023)*, no significant difference was found between the group receiving pulse steroids and the group receiving high dose steroids in terms of mortality and duration of intensive care unit stay. In the study by *Watanabe et al. (2023)*, no significant difference was found between the pulse steroid group and the high dose steroid group in terms of mortality and intensive care unit admission in patients hospitalised in intensive care unit. In our study, no significant difference was observed between the groups in terms of duration of intensive care unit stay. We included patients who received methylprednisolone 80 mg as high dose. In these studies, patients in the high dose group

received dexamethasone 6 mg. This may be the reason why mortality was similar in these studies. In addition, this may have been due to the fact that the number of our patients was smaller compared to these studies. Importantly, the lack of statistically significant differences in ICU stay duration across treatment groups suggests that while pulse steroids may effectively address inflammation and reduce mortality, they do not necessarily expedite the recovery process in terms of hospital liberation. This finding highlights a critical area for further research, as understanding the interplay between inflammation control and recuperation time will be key to optimizing patient management.

Our study has some important limitations. We do not clearly know the duration of exposure to COVID-19 infection. New studies considering the duration of exposure to infection are needed. In our study, we do not have a control group that who did not receive steroid treatment. We included patients with moderate and severe COVID-19. New studies including the control group and patients with mild COVID-19 can be planned. New multicentre studies with a larger number of patients are needed.

CONCLUSION

In conclusion, our study adds to the growing body of evidence that pulse steroid treatment may serve as a more effective strategy in reducing mortality among critically ill COVID-19 patients, compared to high-dose corticosteroid treatments. Future randomized controlled trials are warranted to solidify these findings and optimize treatment protocols that address the complexities of ARDS in the context of COVID-19. Exploring patient characteristics that predict responsiveness to pulse steroids *versus* high-dose regimens may further enhance individualized patient care, ultimately improving survival outcomes in this vulnerable population.

ADDITIONAL INFORMATION AND DECLARATIONS

Funding

The authors received no funding for this work.

Competing Interests

The authors declare there are no competing interests.

Author Contributions

- Huseyin Ali Ozturk conceived and designed the experiments, performed the experiments, analyzed the data, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.
- Fatih Necip Arici conceived and designed the experiments, performed the experiments, analyzed the data, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.

Human Ethics

The following information was supplied relating to ethical approvals (*i.e.*, approving body and any reference numbers): Adana City Training and Research Hospital Ethics Committee approved the study with decision number 2784 dated 17.08.2023.

Data Availability

The following information was supplied regarding data availability:

The raw measurements are available in the Supplementary File.

Supplemental Information

Supplemental information for this article can be found online at http://dx.doi.org/10.7717/peerj.18606#supplemental-information.

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